Process Characterization Plan

A-Mab Drug Substance — Anion Exchange Chromatography (Step 8) — PCP-008

Process Development / Manufacturing Science & Technology (MSAT)

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Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

AEX, anion exchange chromatography; ALCOA+, attributable, legible, contemporaneous, original and accurate, plus complete, consistent, enduring and available; AMV, analytical method validation; ANOVA, analysis of variance; BLA, Biologics License

Application; CCD, central composite design; CEX, cation exchange chromatography; CPP, critical process parameter; CQA, critical quality attribute; CV, column volume; DNA, deoxyribonucleic acid; DoE, design of experiments; DS, drug substance; ELISA, enzyme-linked immunosorbent assay; GPP, general process parameter; HCP, host cell protein; icIEF, imaged capillary isoelectric focusing; IgG1, immunoglobulin G subclass one; KPP, key process parameter; LRF, log reduction factor; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; OLS, ordinary least squares; PAR, proven acceptable range; PD, Process Development; PPQ, process performance qualification; PRESS, prediction error sum of squares; qPCR, quantitative polymerase chain reaction; QA, Quality Assurance; RSM, response surface methodology; SDM, scale-down model; SOP, standard operating procedure; TCID50, fifty percent tissue culture infectious dose; WC-CPP, well-controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

1 Purpose and scope

This plan defines the Stage 1 process characterization study for the anion exchange chromatography step of the A-Mab drug substance process, Step 8 of the purification train. The step is the last chromatographic step in the train and is operated in flow-through mode. Its characterization supports the transfer of the process from the development site to the commercial drug substance site under PTP-001, and it will be executed on a qualified scale-down model of the commercial column (U.S. Food and Drug Administration 2011).

The study covers the 5 process parameters that RA-001 carried forward for this step and the 4 responses measured on the flow-through pool. Of those parameters, 4 will be studied in a multivariate design and 1 one at a time. The commercial process is operated at a bioreactor scale of 15,000 L, and the anion exchange step processes the cation exchange pool derived from a single batch at that scale.

Three activities are outside this plan. Resin packing, sanitization and life cycle are qualified under SOP-2008, and none of them is varied here. Confirmation of the ranges at commercial scale belongs to Stage 2 and will be performed during process performance qualification. The small virus retentive filter is characterized in PCP-009, and no clearance credited to that filter is claimed by this study.

2 Objectives

The study has five objectives.

1. Quantify the effect of each studied parameter on pool host cell protein, on the log reduction factors for xenotropic murine leukaemia virus and minute virus of mice, and on step yield, over the ranges given in §6.1.
2. Define the multivariate operating region within which every governed quality attribute meets its criterion, and state where the boundary of that region lies.
3. Establish a proven acceptable range for each parameter and each attribute it governs, and confirm that the normal operating range falls inside it.
4. Classify each parameter under SOP-4001 from the effects demonstrated in the study and from the capability with which the parameter can be held.
5. Provide the log reduction claim this step contributes to the modular viral clearance assessment, and justify the Stage 2 ranges and this step's part of the control strategy.

The designs in §6 meet the first three objectives. The classification rules in SOP-4001, applied to those results, meet the fourth, and the spiking runs described in §5.1 and §5.4 meet the last. Every objective is reported in PCR-008.

3 Related documents

The documents that scope this study and that consume its output are listed in Table 3.1. RA-001 fixes the parameters and the ranges carried into the study, and PCR-008 will report the executed study against this plan.

Table 3.1: Related documents in the A-Mab characterization package.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-008	Process Characterization Report (Anion Exchange Chromatography (Step 8))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

The controlled procedures and the validated analytical methods the study will follow are listed in Table 3.2. SOP-1001 governs qualification of the scale-down model, SOP-1002 the design and the statistical analysis of the multivariate studies, and SOP-2011 the operation of the step itself.

Table 3.2: Controlled procedures and validated methods for the study.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2011	Operation of the Anion-Exchange Polishing Chromatography Step	SOP
SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3012	Host-Cell Protein ELISA	Method validation

Reference	Title	Type
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation
AMV-3017	XMuLV Infectivity Titre (TCID50)	Method validation
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit operation description and prior knowledge

Anion exchange chromatography is operated in flow-through mode on a resin carrying a quaternary amine ligand. At the load pH the antibody carries a net positive charge, because its isoelectric point lies well above the pH of the load, and it passes through the bed. Species that are negatively charged at that pH bind to the ligand instead. These include residual DNA, the acidic fraction of the host cell protein population, leached Protein A and virus particles, whose surfaces are acidic. The step therefore removes impurities by charge while the product is not retained, and yield in flow-through mode is high.

Binding of an impurity depends on its net charge at the load pH and on how strongly the ionic strength of the buffer shields that charge. Raising the load pH moves more of the host cell protein population below its isoelectric point and into the bound fraction. Raising the conductivity of the equilibration and first wash buffer shields the electrostatic interaction, weakens the binding of the more weakly charged species and lets them through. Load pH and buffer conductivity therefore push the same binding equilibrium in opposite directions, and the clearance of the step is the balance between them. An interaction between the two is expected for that reason, and the design must be able to estimate it free of confounding.

The step also carries the anion exchange contribution to viral safety. Retrovirus-like particles present an acidic surface at the load pH and bind the quaternary amine ligand by the same electrostatic mechanism as the acidic impurities. The parvovirus model has no envelope and is small, and its capsid is acidic at the load pH, so it binds by that mechanism as well. Clearance of both is expected to fall as the load pH drops toward the point at which the particle surface loses its negative charge, and as conductivity rises. Minute virus of mice is not inactivated by low pH, so this step and the virus filter carry the whole cumulative claim for it (International Council for Harmonisation 2023a).

Residual DNA is a polyanion at every pH in the range studied and binds the resin strongly. Its clearance is expected to be robust to the operating parameters, and it is monitored across the step rather than modelled (§5.4). Leached Protein A is acidic and binds the resin while the antibody passes through, so the step is expected to clear the ligand fragments the capture step introduced. Fragments held in complex by the antibody Fc are removed only where the complex dissociates, and no clearance is claimed for that fraction.

The platform anion exchange step has been used to manufacture three previously licensed humanized IgG1 antibodies (X-Mab, Y-Mab and Z-Mab) and through A-Mab clinical supply (CMC Biotech Working Group 2009). Prior product knowledge established the mode of operation, the buffer system and the order of magnitude of the clearance the step delivers, and the mechanism of separation is charge based and does not depend on the fine structure of the antibody. This study therefore confirms and bounds behaviour that is already understood. The ranges in §6.1 are wider than the ranges used in clinical manufacture, so that the edge of the operating region lies inside the data rather than beyond it.

4.2 Quality attributes in scope

The step sets one critical quality attribute, which is listed in Table 4.1. Cumulative clearance of minute virus of mice is a viral safety attribute of very high criticality, and it is the attribute expected to constrain the operating region of this step.

Table 4.1: Critical quality attribute set by the anion exchange step.

CQA	Category	Acceptance	Criticality	Tool #1
Viral clearance — MVM (parvovirus)	viral safety	8.6–20 log10 (cumulative)	VH	20

Further attributes are governed at this step without being set by it, and these are listed in Table 4.2. Host cell protein, residual DNA and leached Protein A are formed upstream and are cleared by more than one step, so the criterion that applies to the anion exchange pool is an in-process limit and not the drug substance specification (§7). Clearance of xenotropic murine leukaemia virus is credited to the low-pH hold, to this step and to the virus filter, and each contributes an independent increment to the cumulative claim (International Council for Harmonisation 2023a).

Table 4.2: Quality attributes governed but not set by the anion exchange step.

CQA	Category	Acceptance	Criticality	Tool #1
Viral clearance — XMuLV (enveloped)	viral safety	16.7–30 log10 (cumulative)	VH	20
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36
Residual DNA	process impurity	0–0.001 ng/dose	VL	6
Leached Protein A	process impurity	0–5 ppm	L-M	16

Criticality was assigned in RA-001 from the impact of the attribute on safety and efficacy combined with the uncertainty in that impact, and attributes scoring high or very high on that combination are treated as critical (CMC Biotech Working Group 2009; International Council for Harmonisation 2023b). Step yield is a process performance attribute rather than a quality attribute. It is measured on every run and reported, and no acceptance criterion is applied to it at this step.

4.3 Risk-based prioritization of parameters

RA-001 ranked every parameter of the step on its potential impact on a critical quality attribute and on its potential to interact with other parameters, and the ranking decided the study type. Parameters ranked high on both counts were assigned to the multivariate design. Parameters ranked high on impact but low on interaction were assigned to a univariate assessment. Parameters held to a platform specification with no credible path to a quality attribute were not studied, and their ranges are carried from platform knowledge and from the modular claims established on prior products.

The outcome for this step is 4 parameters in the multivariate design and 1 assessed one at a time (§6.1). Load pH, the conductivity of the equilibration and first wash buffer and the conductivity of the load were ranked highest, because they act on the same binding equilibrium and their interaction is the route by which impurity breakthrough occurs. Protein load was carried into the design because the impurity mass loaded onto the column scales with it, and the ligand has a finite capacity for the species it binds. Operating flow rate was assigned to a univariate assessment, because binding of the small, highly charged impurities is fast relative to the residence time at every flow rate in the range.

The material loaded onto the step is the cation exchange pool, whose attributes are controlled by the preceding step under PCP-007. This study varies the parameters of the anion exchange step and does not vary the quality of its load. The load material will be qualified against the release profile of that pool before execution (§7).

5 Materials and methods

5.1 Scale-down model and its qualification

The study will be executed on a scale-down model of the commercial anion exchange column, qualified under SOP-1001. The model keeps the bed height, the linear flow velocity, the protein load in grams of antibody per litre of resin, and the equilibration, load, wash and chase volumes expressed in column volumes. Scaling on these variables holds the residence time and the impurity mass per unit of ligand equal to the commercial step, and those are the two quantities on which the binding equilibrium depends. The resin, the buffers and the packing method are the same as at commercial scale.

Qualification will compare the model against commercial-equivalent runs on step yield, pool host cell protein, residual DNA, leached Protein A, the pressure across the bed and the shape of the flow-through peak. The model will be accepted where no statistically significant difference in these attributes is found between the scales. Where a difference is found, the model should be modified and requalified before the characterization runs begin, and no characterization run may be executed on an unqualified model (U.S. Food and Drug Administration 2011). The qualification record is held with the study data and is cited in PCR-008.

The viral clearance runs will be executed on the same scale-down model in a dedicated containment laboratory, as a small-scale spiking study (International Council for Harmonisation 2023a). The spiking model will be qualified separately, because adding a virus spike to the load changes the load itself. That qualification will confirm that the spiked load matches the unspiked load in protein concentration, in pH and in conductivity, and that the spike volume stays below the fraction at which those attributes move. Spiked and unspiked runs will be executed on matched load material, so that the clearance results and the impurity results describe the same step.

The centre point of each design will be executed in replicate. The replicates estimate pure error, and pure error is the reference against which lack of fit is tested (§5.5). A replicate is a full independent execution of the step and not a repeat measurement of one pool, because a repeat injection estimates assay variability and lack of fit must be judged against the variability of the process.

5.2 Operation

The step will be operated as it is operated at commercial scale, under SOP-2011. The column is equilibrated with the equilibration buffer, loaded with the cation exchange pool adjusted to the target pH and conductivity, and chased with the first wash buffer. The flow-through and the chase are collected as a single pool from the start of the load to the end of the chase. The pool is filtered and held for the virus filtration step, and the hold conditions are those of the commercial process.

Resin, buffers and column packing are controlled to platform specification. Each resin lot will be released against its certificate of analysis and packed and tested under SOP-2008, and a packed column will be accepted on plate count and peak asymmetry before use. Buffer lots will be prepared and released under the site buffer preparation procedure and used within their assigned expiry. These inputs are identity and quality controlled, and they are not characterized in this study.

The chromatography system will be operated in a temperature-controlled chamber under calibration and change control. The system is identified in Table 5.1, and its calibration status will be verified before the first run and confirmed at the end of the execution period.

Table 5.1: Instrumented system used for the anion exchange characterization runs.

Equipment	Description	Calibration status	Next calibration due
EQ-CHR-118	temperature-controlled chromatography chamber	in calibration at execution	2026-05-30

5.3 Analytical methods

Every response will be measured by a validated method, and the methods are listed with their validation references in Table 3.2. Pool host cell protein will be measured by ELISA (AMV-3012) and reported in nanograms per milligram of antibody. Residual DNA will be measured by qPCR (AMV-3014) and leached Protein A by ELISA (AMV-3016). Infectivity of xenotropic murine leukaemia virus will be measured by TCID50 (AMV-3017) and infectivity of minute virus of mice by TCID50 with a qPCR confirmation (AMV-3018). The charge variant profile of the load will be measured by icIEF (AMV-3013) and compared with the release profile of the cation exchange pool (§7).

An effect smaller than the variability of the method that measures it cannot be separated from that variability. The precision of each assay therefore bounds the smallest effect the study can resolve on its response. Method variability enters the residual variance of the fitted models and is not separated from process variability by this design. An apparent effect on a log reduction factor that is small relative to the variability of the infectivity assay will be attributed to the assay rather than to the step. Performance characteristics of each method are held in the validation reports named in Table 3.2 and are not repeated here.

5.4 Sampling plan

Samples will be taken from the load and from the flow-through pool of every run. The load sample supports the material qualification in §7 and provides the input concentration against which clearance is calculated. The pool sample carries the responses.

The screening design includes 3 centre point replicates and the response surface design includes 4. Each replicate is executed as a separate run from a separate load aliquot, for the reason given in §5.1. Load and pool samples will be retained frozen so that a response can be re-tested if an assay result is invalidated.

Viral clearance runs will be sampled at the spiked load and at the pool. The log reduction factor will be calculated from the total infectivity in the load and the total infectivity in the pool. Cytotoxicity and interference controls will be run for each matrix, and the detection limit of the assay will be reported with any result that reaches it (International Council for Harmonisation 2023a).

Residual DNA and leached Protein A will be measured on the pool of every run in the response surface design. They are monitored to confirm the clearance the mechanism in §4.1 predicts, and they are not fitted as responses of the design. Where either attribute moves with a parameter, the observation will be reported and carried into the risk assessment rather than used to set a range.

5.5 Statistical methods

Each parameter will be coded so that the set-point is 0 and the edges of the characterization range are ± 1. Coding places the parameters on one scale, so that an effect estimated in coded units is comparable across parameters measured in different physical units. Effects will be converted back to natural units when a range is reported.

The screening data will be analysed by ordinary least squares with every main effect and every two-factor interaction in the model. An effect will be reported as twice the coded coefficient, which is the change in the response across the full characterization range of the parameter. Terms significant at $p < 0.05$ will be identified as active. The screening model identifies which parameters act on which response, and it is not the model from which an operating region is derived.

The response surface data will be analysed with a full quadratic model in the 4 multivariate parameters, fitted on the coded factors. This is the predictive model of the study. The operating region, the proven acceptable ranges and the capability estimate will all be derived from it, and PCR-008 will report the coefficients, the analysis of variance and the diagnostics for each response.

Model adequacy will be judged on four things. The coefficient of determination adjusted for the number of terms and the predicted coefficient of determination computed from the prediction error sum of squares will be reported together, because a large gap between them indicates a model that fits the data it was built on and predicts poorly away from them. The lack of fit sum of squares will be tested against pure error from the centre point replicates by an F test. Residuals will be inspected against the predicted values and on a normal quantile plot. No numeric threshold for the coefficient of determination is fixed in advance, because what an adequate fit means depends on the size of the effects the model must resolve relative to the pure error of the step, and the decision rules that do apply are stated in §7.

A parameter with no active term in either model will be described as having no detectable effect on that response over the range studied. The range studied bounds that statement, and it will be stated with the range attached.

Capability at commercial scale will be estimated by Monte-Carlo simulation of 2,000 batches, with every parameter drawn from within its normal operating range and propagated through the fitted models and their residual variance. A one-sided capability index will be reported for each governed attribute against the criterion that applies to it. The estimate carries the residual variance of the fitted models and the spread of the parameters inside their normal operating ranges, and it carries no source of commercial variability that the scale-down model does not reproduce. Confirmation at commercial scale is a Stage 2 activity (U.S. Food and Drug Administration 2011).

6 Study design

6.1 Factors, ranges and study type

Table 6.1 lists the parameters, their set-points, the range that will be studied, the normal operating range and the study type assigned in RA-001.

Table 6.1: Parameters, ranges and study type for the anion exchange characterization.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Load pH	pH	7.5	7.2–7.8	7.35–7.65	multivariate
Equil/Wash-1 conductivity	mS/cm	2.6	1.6–3.6	2.1–3.1	multivariate
Load conductivity	mS/cm	5.5	3–8	4–7	multivariate
Protein load	g/L resin	200	50–300	120–280	multivariate
Operating flow rate	cm/hr	300	150–450	200–400	univariate

The range studied is wider than the normal operating range for every parameter. A study confined to the normal operating range would place the edge of the operating region outside the data, and a proven acceptable range may only be claimed over settings that were actually run (CMC Biotech Working Group 2009).

The load pH range is bounded above by the pH at which a fraction of the antibody would begin to bind and yield would fall. The range in Table 6.1 sits below that pH, so product loss is not expected anywhere in the design. The conductivity of the equilibration and first wash buffer is set by buffer preparation and is the more closely controlled of the two conductivities, and its range reflects the spread that buffer preparation and column equilibration can produce. The conductivity of the load is set by the cation exchange pool and by its adjustment, so its range reflects what the preceding step delivers rather than a choice made at this step. The protein load range extends well above the set-point, because the impurity mass carried with the antibody scales with the load and the ligand has a finite capacity for the bound species.

Table 6.2 gives the letter codes used for the multivariate parameters in the design matrices in Appendix A and Appendix B, and the same codes will be used for the model terms reported in PCR-008.

Table 6.2: Coded factor legend for the multivariate designs.

Code	Factor	Unit	Studied in
A	Load pH	pH	screening + RSM
B	Equil/Wash-1 conductivity	mS/cm	screening + RSM
C	Load conductivity	mS/cm	screening + RSM
D	Protein load	g/L resin	screening + RSM

6.2 Screening design

The 4 multivariate parameters will be studied in a two-level full factorial design of 16 runs, augmented with 3 centre points, for 19 runs in total. A full factorial in these parameters estimates every main effect and every two-factor interaction without confounding, which the interaction between load pH and wash conductivity requires. The centre points estimate pure error and give a test for curvature, and curvature at the centre indicates that a quadratic model is needed.

Runs will be executed in randomized order within one resin lot and one load lot. Randomization protects the estimates against drift in the load material, in the resin or in the assay over the execution period. Where the design cannot be completed within one load lot, the lots should be treated as a block and the block term reported with the effects. The planned matrix is given in Appendix A.

6.3 Response-surface design

The same parameters will be studied in a face-centred central composite design of 28 runs. The design adds 8 axial runs at the faces of the cube to 16 factorial runs and 4 centre points. Axial points placed at the faces keep every run inside the characterization range, which an axial distance beyond the faces would not.

The quadratic terms this design supports are needed because the net charge on a protein or a capsid surface changes most steeply near its isoelectric point, so clearance rises more sharply over the part of the load pH range that lies nearest the isoelectric point of the bound species than over the rest of it. A design with two levels in each parameter would fit that curve as a straight line and would put the edge of the operating region at the wrong pH. The planned matrix is given in Appendix B.

The response surface design will be executed on load material qualified under the criteria in §7. Where it cannot be executed on the same load lot as the screening design, the second lot will be compared with the first on charge variant profile, host cell protein and protein concentration before the runs begin.

6.4 Univariate assessment

Operating flow rate will be assessed one at a time over the range in Table 6.1, with every other parameter held at its set-point. The low, the centre and the high setting of the range will be executed, and pool host cell protein and step yield will be measured on each. A viral clearance run at the high setting will be included, because residence time is the mechanism by which flow rate could act on clearance and the high setting is the shortest residence time in the range.

A univariate assessment cannot resolve an interaction between flow rate and another parameter, and the claim it supports is bounded by the assumption that the effect of flow rate does not depend on where the other parameters sit. Binding of the small, highly charged impurities is fast relative to the residence time at every flow rate in the range, so residence time is not expected to limit clearance anywhere in the design. Should the assessment show an effect of flow rate on any response, the parameter will be carried into a multivariate design before its range is claimed.

The result will support the range for flow rate as a proven acceptable range, and it will not enter the response surface model.

7 Acceptance and decision criteria

This section states the rules by which the study will be judged. PCR-008 will report the executed study against each of them, and a criterion that is not met will be dispositioned as a deviation rather than revised after the fact.

The scale-down model will be accepted where the qualification in §5.1 finds no statistically significant difference between the model and commercial-equivalent operation on the compared attributes. An unqualified model releases no characterization run.

Load material will be accepted as representative where its charge variant profile by icIEF, its host cell protein content, its protein concentration, its pH and its conductivity fall within the release profile of the cation exchange pool. Load material outside that profile is not representative of routine operation, and a design executed on it will be invalidated for the definition of the operating region and re-executed in full on a qualified load. Data from an invalidated execution will be retained and may be used to investigate and confirm the cause, and they will not be used to set a range, a classification or a clearance claim.

A model will be accepted as adequate where the lack of fit test in §5.5 returns $p \geq 0.05$ against pure error and the residual diagnostics show no structure. Where lack of fit is significant, the model may still be used for prediction if the residual standard error is small relative to the margin between the predicted response and its limit, and the justification for using it will be recorded in PCR-008. A model that is neither adequate nor justified supports no range, and the affected response will be reported without a predictive model.

Each governed response will be judged against the criterion that applies to it at this step. Pool host cell protein will be judged against the in-process limit of 21.7 ng/mg, which is carried back from the drug substance criterion of 100 ng/mg through the clearance the downstream steps deliver and divided by an assurance margin. Clearance of xenotropic murine leukaemia virus will be judged against a step floor of 3.78 log10, and clearance of minute virus of mice against a step floor of 3.28 log10. Each floor is the cumulative requirement for that virus, 16.7 and 8.6 log10 respectively, less the clearance credited to the other contributing steps. Step yield carries no acceptance criterion at this step and will be reported as a performance attribute.

The multivariate operating region will be declared over the part of the characterized region in which every governed response meets its criterion simultaneously. The normal operating range of every parameter must fall inside that region. Where it does not, the normal operating range will be narrowed in the control strategy and the change will be carried into PCMR-001.

Each parameter will be classified under SOP-4001 from the effect demonstrated on a critical quality attribute and from the capability with which the parameter is held. A parameter that acts on a critical quality attribute and is held well inside its acceptable range is a well-controlled critical process parameter. A parameter that acts on process performance without acting on a quality attribute is a key process parameter. A parameter with no demonstrated effect on either is a general process parameter, and the range studied bounds that classification.

The study will be judged complete when the scale-down model is qualified, every planned run is executed or dispositioned, every response has an adequate model or a stated reason for the absence of one, the proven acceptable ranges bracket the normal operating ranges, and the clearance measured for both model viruses meets the step floors given above.

8 Proven acceptable ranges (planned analysis)

A proven acceptable range is the range of one parameter over which the attribute it governs stays within its criterion, with the other parameters held or varied as the analysis states. Two analyses will be run for every pair of a parameter and an attribute it governs, and both will be reported.

The first holds the other parameters at their set-points and evaluates the fitted response surface model across the parameter's characterization range. The second varies the other parameters within their normal operating ranges by Monte-Carlo simulation of the fitted model, including its residual variance, and reads the range over which the predictive interval of the response stays within the criterion. The interval used is two sided and covers the central 95 % of the simulated distribution. The normal operating ranges in the control strategy will be set from the second analysis, and the first will be reported beside it.

The analysis will be evaluated on a grid of 81 settings across the characterization range, with 2,000 Monte-Carlo draws at each setting, under a fixed random seed recorded with the analysis (§9). A range will be reported only where the acceptable settings form one contiguous interval containing the set-point. Where they do not, the discontinuity will be reported as it stands and no range will be claimed for that pair.

Each range will be reported in the natural units of the parameter together with the criterion it was measured against (§7). For each governed attribute the report will also carry a plot of the response against its governing parameter, with the acceptable region shaded green and the set-point and the normal operating range marked. The governing parameter is the multivariate parameter with the largest main effect on that attribute in the response surface model.

The univariate proven acceptable range for operating flow rate will be derived from the assessment in §6.4 and reported beside the multivariate ranges, with the assumption stated in §6.4 attached to it. Where a proven acceptable range does not contain the whole normal operating range, the normal operating range will be narrowed rather than the criterion relaxed.

9 Data management and integrity

Data generated under this plan will be attributable, legible, contemporaneous, original and accurate, and the records will be complete, consistent, enduring and available. Raw data will be captured in the chromatography control system, in the laboratory information system and on the analytical instruments that generate them, with audit trails enabled.

Each run will carry the identity of the resin lot, the packed column, the load lot, the buffer lots, the instrument and the analyst. Chromatograms and instrument files will be retained in their original electronic form. A run that is aborted or that departs from SOP-2011 will be recorded with the reason, and the disposition of its data will be decided before the analysis begins.

The statistical analysis will be executed by a scripted analysis held under version control, and the script, its inputs and its outputs will be retained with the study record. The Monte-Carlo analyses use a fixed random seed of 20240724, so that a reported range can be reproduced exactly from the same data. Any change to the analysis after the first execution will be recorded with its reason.

Data will be reviewed by a second qualified person against the raw records before the analysis is run, and the review will be documented. Deviations from this plan will be recorded and dispositioned in PCR-008 under the site deviation procedure, with an impact assessment for each.

10 Roles and responsibilities

Responsibilities are assigned by function in Table 10.1. The functions are those that approve this plan, with the addition of virology, which executes the spiking runs in containment.

Table 10.1: Responsibilities by function.

Function	Responsibility
Process Development / MSAT	Owns this plan, qualifies the scale-down model, executes the characterization runs and authors PCR-008
Analytical Development	Performs the validated assays, maintains their validation status and releases the analytical data
Virology	Executes the spiking runs, calculates the log reduction factors and reports the assay controls
Statistics	Approves the designs, fits the models, performs the range and capability analyses and reviews the classification
Manufacturing Science (receiving site)	Reviews the proposed ranges against the control capability of the commercial equipment
Quality Assurance	Approves this plan and PCR-008 and dispositions deviations

The receiving site reviews the ranges before they are fixed, because a range that the commercial equipment cannot hold is of no use in a control strategy. Statistics approves the designs before execution rather than after, since a design that cannot estimate the interaction between load pH and wash conductivity cannot be repaired by analysis.

11 Deliverables and schedule

The deliverable of this plan is PCR-008, the process characterization report for the anion exchange step. It will report the executed designs, the fitted models and their adequacy, the operating region, the proven acceptable ranges, the classification of each parameter, the clearance claimed for the step and the contribution this step makes to the control strategy. PCR-008 rolls up into PCMR-001, which consolidates the cumulative viral clearance claim across the contributing steps.

The study will run in four phases. The scale-down model is qualified first, because no characterization run may start on an unqualified model. The screening design is executed next, then the response surface design, then the univariate assessment. The viral clearance runs may run in parallel with the univariate assessment, because they use a separate laboratory and separate load aliquots.

No calendar dates are committed in this plan. The sequence above is the dependency between the phases, and the campaign schedule is held in PTP-001.

12 Risks and assumptions

The scale-down model may not represent the commercial column. Qualification under SOP-1001 addresses this risk, and every claim made from the model is bounded by that qualification (§5.1). Design spaces are not confirmed at scale at the edges of the ranges, and confirmation at commercial scale is a Stage 2 activity.

The load material may not be representative of the routine cation exchange pool. An extended neutralized hold of that pool raises its acidic charge variant burden, which changes the charge distribution of the species this step is asked to bind and can therefore change what the step clears. Load material will be qualified before execution against the criteria in §7, and a design executed on non-representative material will be re-executed.

The virus spike may perturb the load. The spike volume will be bounded and the spiked load compared with the unspiked load, as §5.1 describes.

Material supply may constrain the execution. The designs together require a large volume of cation exchange pool, and pooling several cation exchange batches to obtain it would blur the load attributes the study holds constant. Load lots will therefore be scheduled against the runs before execution begins.

The study assumes that platform knowledge from X-Mab, Y-Mab and Z-Mab transfers to A-Mab at this step, because the separation is charge based and the platform buffer system is unchanged. It assumes the resin performs within its qualified life cycle over the execution period under SOP-2008. It assumes that parameters not carried into the study are held to platform specification, and an excursion in any of them will be recorded as a deviation and assessed for impact in PCR-008.

13 Approvals

This plan will be executed after approval by the functions listed in the approval record in the front matter. Approval by Quality Assurance releases the study for execution. A change to this plan after approval will be made by revising the plan and will be approved by the same functions before the affected runs are executed.

14 Appendices

The matrices below are the planned designs. No result is shown, and the executed matrices with their measured responses will be reported in the appendices of PCR-008.

14.1 Appendix A — Planned screening design matrix

The runs are given in coded units, where the set-point is 0 and the edges of the characterization range are ±1. The letter codes are given in Table 6.2 and the natural settings follow from the ranges in Table 6.1. Run order will be randomized at execution.

Table 14.1: Planned screening design matrix, coded units.

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	center	0	0	0	0
18	center	0	0	0	0
19	center	0	0	0	0

14.2 Appendix B — Planned response-surface design matrix

The face-centred central composite design is given in the same coded units. The axial runs sit at the faces of the cube, so no run lies outside the characterization range.

Table 14.2: Planned response-surface design matrix, coded units.

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	axial	-1	0	0	0
18	axial	1	0	0	0
19	axial	0	-1	0	0
20	axial	0	1	0	0
21	axial	0	0	-1	0
22	axial	0	0	1	0
23	axial	0	0	0	-1
24	axial	0	0	0	1
25	center	0	0	0	0
26	center	0	0	0	0
27	center	0	0	0	0
28	center	0	0	0	0

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